

distribution_checks_bb

August 6, 2024

1 Initial Data Analysis and Distribution Checks

```
[ ]: # Data analysis
import pandas as pd
import numpy as np

# Plotting and Correlation Maths
import seaborn as sns
import scipy as sci

# Simple model development
import sklearn as sk

import matplotlib.pyplot as plt

import plotly as px

import os
```

2 Data import and initial check

```
[ ]: data = pd.read_csv('../data/processed/cleaned_data.csv') # Clean data runs
# data = pd.read_csv('../data/interim/merged_df.csv') # old runs
```

```
[ ]: data
```

```
[ ]: 
```

	tempmax	tempmin	temp	feelslikemax	feelslikemin	feelslike	dew	\
0	26.2	-4.2	10.1	18.1	-16.6	0.6	-3.8	
1	35.8	28.0	32.5	34.7	24.1	26.8	26.6	
2	15.3	2.5	9.2	14.6	-6.0	3.5	0.5	
3	15.3	2.5	9.2	14.6	-6.0	3.5	0.5	
4	15.3	2.5	9.2	14.6	-6.0	3.5	0.5	
...	...	...	...	...	...	...	...	
3679	68.8	58.9	61.6	68.8	58.9	61.6	56.7	
3680	59.4	46.8	50.8	59.4	39.8	46.5	46.3	
3681	59.2	44.5	51.1	59.2	37.4	47.9	33.0	

3682	81.1	45.0	64.1	79.9	39.9	62.9	39.5
3683	81.1	45.0	64.1	79.9	39.9	62.9	39.5

	humidity	precip	precipcover	...	preciptype_freezingrain	\
0	57.2	0.000	0.00	...	0	
1	78.9	0.000	0.00	...	0	
2	68.0	0.000	0.00	...	0	
3	68.0	0.000	0.00	...	0	
4	68.0	0.000	0.00	...	0	
...	...	...	...	...	...	
3679	84.8	1.526	83.33	...	0	
3680	84.5	0.266	83.33	...	0	
3681	51.5	0.019	12.50	...	0	
3682	42.6	0.001	4.17	...	0	
3683	42.6	0.001	4.17	...	0	

	preciptype_freezingrain,snow	preciptype_freezingrain,snow,ice	\
0	0	0	
1	0	0	
2	0	0	
3	0	0	
4	0	0	
...	...	...	
3679	0	0	
3680	0	0	
3681	0	0	
3682	0	0	
3683	0	0	

	preciptype_none	preciptype_rain	preciptype_rain,freezingrain	\
0	1	0	0	
1	1	0	0	
2	1	0	0	
3	1	0	0	
4	1	0	0	
...	...	...	...	
3679	0	1	0	
3680	0	1	0	
3681	0	1	0	
3682	0	1	0	
3683	0	1	0	

	preciptype_rain,freezingrain,snow	preciptype_rain,snow	\
0	0	0	
1	0	0	
2	0	0	
3	0	0	

```

4                0                0
...              ...              ...
3679             0                0
3680             0                0
3681             0                0
3682             0                0
3683             0                0

```

```

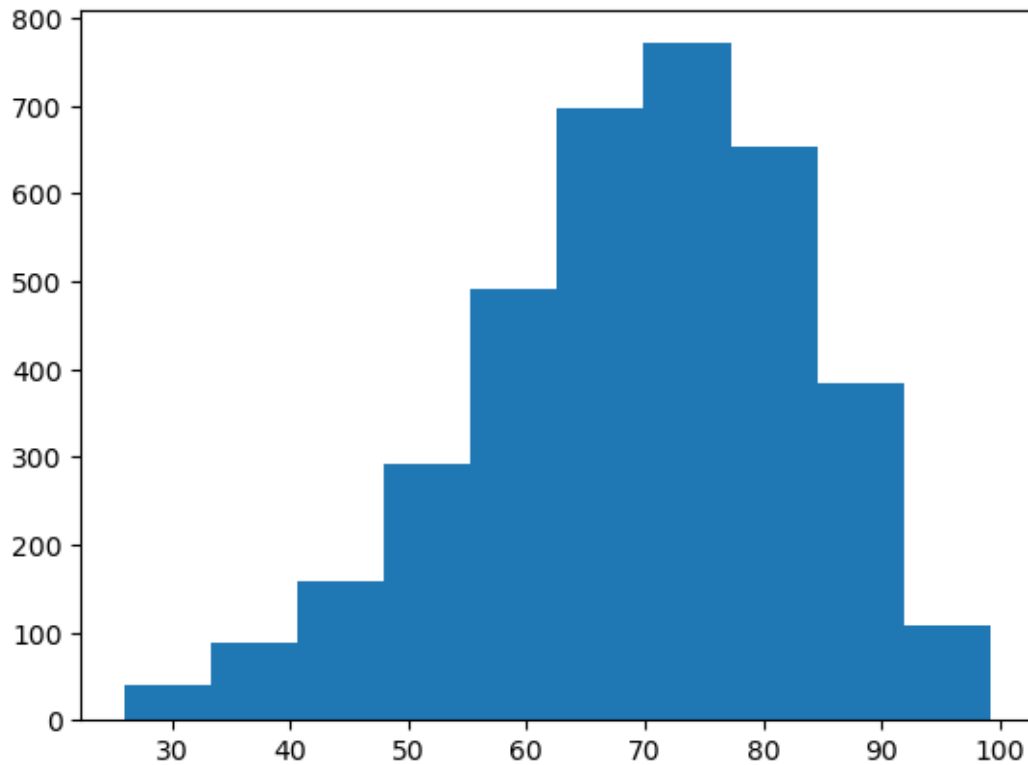
      precipitype_rain,snow,ice  precipitype_snow
0                0                0
1                0                0
2                0                0
3                0                0
4                0                0
...              ...              ...
3679             0                0
3680             0                0
3681             0                0
3682             0                0
3683             0                0

```

[3684 rows x 26 columns]

3 Simple Histogram if we need more of them in the future

```
[ ]: plt.hist(data['humidity']);
```



4 How many NAs exist

```
[ ]: # A DataFrame where values have been assigned True if they are NA
nas = data.isna()
# Sum to find out how many values are missing
print("The number of missing values for each column:")
nas.sum()
```

The number of missing values for each column:

```
[ ]: tempmax      0
    tempmin      0
    temp         0
    feelslikemax  0
    feelslikemin  0
    feelslike     0
    dew           0
    humidity      0
    precip        0
    precipcover   0
    snow          0
```

```

snowdepth          0
cloudcover          0
uvindex             0
severerisk          0
num_date            0
preciptype_freezingrain 0
preciptype_freezingrain,snow 0
preciptype_freezingrain,snow,ice 0
preciptype_none     0
preciptype_rain      0
preciptype_rain,freezingrain 0
preciptype_rain,freezingrain,snow 0
preciptype_rain,snow 0
preciptype_rain,snow,ice 0
preciptype_snow      0
dtype: int64

```

```
[ ]: nas
```

```

[ ]:      tempmax  tempmin   temp  feelslikemax  feelslikemin  feelslike  dew  \
0      False    False  False        False        False    False  False
1      False    False  False        False        False    False  False
2      False    False  False        False        False    False  False
3      False    False  False        False        False    False  False
4      False    False  False        False        False    False  False
...      ...      ...      ...      ...      ...      ...      ...
3679   False    False  False        False        False    False  False
3680   False    False  False        False        False    False  False
3681   False    False  False        False        False    False  False
3682   False    False  False        False        False    False  False
3683   False    False  False        False        False    False  False

```

```

      humidity  precip  precipcover  ...  preciptype_freezingrain  \
0      False    False        False  ...                          False
1      False    False        False  ...                          False
2      False    False        False  ...                          False
3      False    False        False  ...                          False
4      False    False        False  ...                          False
...      ...      ...      ...      ...      ...
3679   False    False        False  ...                          False
3680   False    False        False  ...                          False
3681   False    False        False  ...                          False
3682   False    False        False  ...                          False
3683   False    False        False  ...                          False

```

```

      preciptype_freezingrain,snow  preciptype_freezingrain,snow,ice  \
0                                False                                False

```

1	False	False
2	False	False
3	False	False
4	False	False
...	...	...
3679	False	False
3680	False	False
3681	False	False
3682	False	False
3683	False	False

	preciptype_none	preciptype_rain	preciptype_rain,freezingrain \
0	False	False	False
1	False	False	False
2	False	False	False
3	False	False	False
4	False	False	False
...	...	...	...
3679	False	False	False
3680	False	False	False
3681	False	False	False
3682	False	False	False
3683	False	False	False

	preciptype_rain,freezingrain,snow	preciptype_rain,snow \
0	False	False
1	False	False
2	False	False
3	False	False
4	False	False
...	...	...
3679	False	False
3680	False	False
3681	False	False
3682	False	False
3683	False	False

	preciptype_rain,snow,ice	preciptype_snow
0	False	False
1	False	False
2	False	False
3	False	False
4	False	False
...	...	...
3679	False	False
3680	False	False
3681	False	False

```

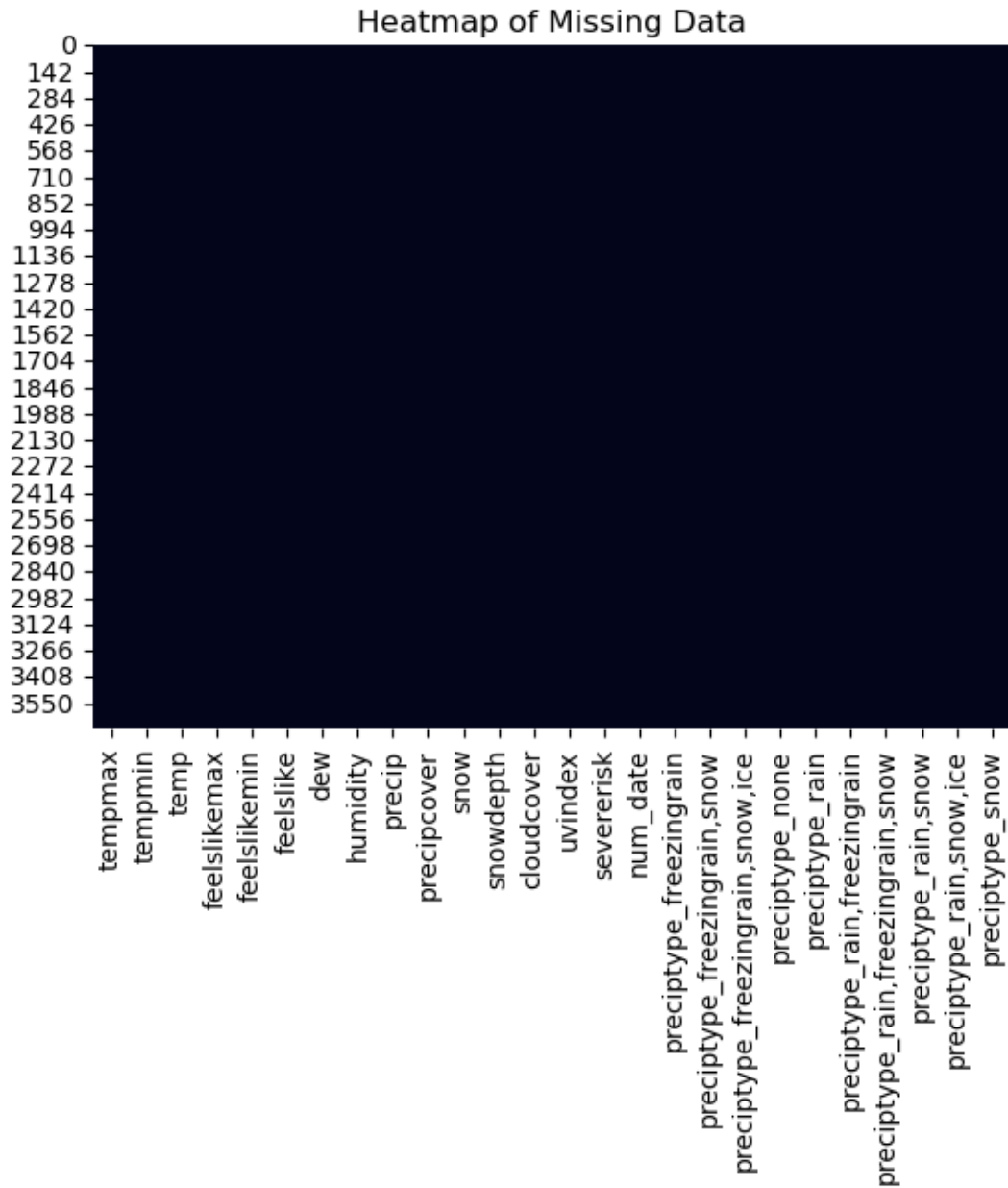
3682                False                False
3683                False                False

```

```
[3684 rows x 26 columns]
```

```
[ ]: sns.heatmap(nas, cbar=False)
plt.title('Heatmap of Missing Data')
```

```
[ ]: Text(0.5, 1.0, 'Heatmap of Missing Data')
```



5 Basic Statistics on features

```
[ ]: cols = data.columns
data[cols[0:12]].describe()
```

```
[ ]:      tempmax      tempmin      temp  feelslikemax  feelslikemin \
count  3684.000000  3684.000000  3684.000000  3684.000000  3684.000000
mean    63.125923   46.715852   54.678067    62.242047    43.368784
std     17.823250   16.136710   16.628591    20.066826    19.642101
min     11.400000   -5.000000    3.800000     2.700000   -30.400000
25%     48.500000   32.800000   40.500000    45.200000    26.200000
50%     66.800000   47.500000   56.400000    66.800000    45.500000
75%     78.600000   61.625000   69.700000    78.600000    61.625000
max     93.700000   75.500000   83.300000   100.600000    75.500000

      feelslike      dew      humidity      precip  precipcover \
count  3684.000000  3684.000000  3684.000000  3684.000000  3684.000000
mean    52.546634   43.638219   69.272693     0.127880    15.574254
std     19.497281   17.188931   13.847836     0.296584    21.785590
min    -13.200000   -7.100000   26.000000     0.000000     0.000000
25%     35.700000   29.100000   60.075000     0.000000     0.000000
50%     56.000000   44.200000   70.450000     0.005000     4.170000
75%     69.700000   59.300000   79.700000     0.110500    25.000000
max     87.200000   74.500000   99.200000     3.922000   100.000000

      snow      snowdepth
count  3684.000000  3684.000000
mean     0.037514    0.051113
std     0.192583    0.188385
min      0.000000    0.000000
25%      0.000000    0.000000
50%      0.000000    0.000000
75%      0.000000    0.000000
max      2.900000    2.400000
```

```
[ ]: data[cols[12:]].describe()
```

```
[ ]:      cloudcover      uvindex      severerisk      num_date \
count  3684.000000  3684.000000  3684.000000  3684.000000
mean    55.168404    4.762215    5.054289   171.845820
std     29.140185    2.430109   10.737648   101.320521
min      0.000000    0.000000    0.000000     1.000000
25%     30.175000    3.000000    0.000000    88.750000
50%     55.900000    5.000000    0.000000   167.000000
75%     80.700000    7.000000   10.000000   254.000000
max    100.000000   10.000000   100.000000   366.000000
```

	preciptype_freezingrain	preciptype_freezingrain,snow \
count	3684.000000	3684.000000
mean	0.002443	0.002714
std	0.049373	0.052037
min	0.000000	0.000000
25%	0.000000	0.000000
50%	0.000000	0.000000
75%	0.000000	0.000000
max	1.000000	1.000000

	preciptype_freezingrain,snow,ice	preciptype_none	preciptype_rain \
count	3684.000000	3684.000000	3684.000000
mean	0.001086	0.372693	0.478013
std	0.032938	0.483587	0.499584
min	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000
75%	0.000000	1.000000	1.000000
max	1.000000	1.000000	1.000000

	preciptype_rain,freezingrain	preciptype_rain,freezingrain,snow \
count	3684.000000	3684.000000
mean	0.003257	0.001086
std	0.056988	0.032938
min	0.000000	0.000000
25%	0.000000	0.000000
50%	0.000000	0.000000
75%	0.000000	0.000000
max	1.000000	1.000000

	preciptype_rain,snow	preciptype_rain,snow,ice	preciptype_snow
count	3684.000000	3684.000000	3684.000000
mean	0.102877	0.001629	0.034202
std	0.303840	0.040329	0.181772
min	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000
75%	0.000000	0.000000	0.000000
max	1.000000	1.000000	1.000000

```
[ ]: stats = data.describe()
stats
```

```
[ ]:
count    tempmax    tempmin    temp    feelslikemax    feelslikemin \
count    3684.000000    3684.000000    3684.000000    3684.000000    3684.000000
mean      63.125923    46.715852    54.678067    62.242047    43.368784
std       17.823250    16.136710    16.628591    20.066826    19.642101
```

min	11.400000	-5.000000	3.800000	2.700000	-30.400000
25%	48.500000	32.800000	40.500000	45.200000	26.200000
50%	66.800000	47.500000	56.400000	66.800000	45.500000
75%	78.600000	61.625000	69.700000	78.600000	61.625000
max	93.700000	75.500000	83.300000	100.600000	75.500000

	feelslike	dew	humidity	precip	precipcover	...	\
count	3684.000000	3684.000000	3684.000000	3684.000000	3684.000000	...	
mean	52.546634	43.638219	69.272693	0.127880	15.574254	...	
std	19.497281	17.188931	13.847836	0.296584	21.785590	...	
min	-13.200000	-7.100000	26.000000	0.000000	0.000000	...	
25%	35.700000	29.100000	60.075000	0.000000	0.000000	...	
50%	56.000000	44.200000	70.450000	0.005000	4.170000	...	
75%	69.700000	59.300000	79.700000	0.110500	25.000000	...	
max	87.200000	74.500000	99.200000	3.922000	100.000000	...	

	preciptype_freezingrain	preciptype_freezingrain,snow	\
count	3684.000000	3684.000000	
mean	0.002443	0.002714	
std	0.049373	0.052037	
min	0.000000	0.000000	
25%	0.000000	0.000000	
50%	0.000000	0.000000	
75%	0.000000	0.000000	
max	1.000000	1.000000	

	preciptype_freezingrain,snow,ice	preciptype_none	preciptype_rain	\
count	3684.000000	3684.000000	3684.000000	
mean	0.001086	0.372693	0.478013	
std	0.032938	0.483587	0.499584	
min	0.000000	0.000000	0.000000	
25%	0.000000	0.000000	0.000000	
50%	0.000000	0.000000	0.000000	
75%	0.000000	1.000000	1.000000	
max	1.000000	1.000000	1.000000	

	preciptype_rain,freezingrain	preciptype_rain,freezingrain,snow	\
count	3684.000000	3684.000000	
mean	0.003257	0.001086	
std	0.056988	0.032938	
min	0.000000	0.000000	
25%	0.000000	0.000000	
50%	0.000000	0.000000	
75%	0.000000	0.000000	
max	1.000000	1.000000	

preciptype_rain,snow	preciptype_rain,snow,ice	preciptype_snow
----------------------	--------------------------	-----------------

count	3684.000000	3684.000000	3684.000000
mean	0.102877	0.001629	0.034202
std	0.303840	0.040329	0.181772
min	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000
75%	0.000000	0.000000	0.000000
max	1.000000	1.000000	1.000000

[8 rows x 26 columns]

```
[ ]: stats.to_csv('../data/processed/fivenumbersummary.csv')
```

```
[ ]: for col in cols:
      print(data[col].value_counts())
```

tempmax

```
78.4    28
79.6    26
57.8    26
61.3    23
75.7    21
```

..

```
25.1     1
24.2     1
90.4     1
59.0     1
43.6     1
```

Name: count, Length: 590, dtype: int64

tempmin

```
64.9    36
56.3    27
31.9    26
66.8    24
38.9    23
```

..

```
39.5     1
12.8     1
45.5     1
57.5     1
19.7     1
```

Name: count, Length: 550, dtype: int64

temp

```
64.6    25
71.2    22
71.9    21
71.0    21
70.8    20
```

```

    ..
78.7    1
46.9    1
28.0    1
17.1    1
61.6    1
Name: count, Length: 579, dtype: int64
feelslikemax
79.6    31
78.4    28
57.8    26
61.3    23
76.6    21
    ..
45.8    1
60.5    1
22.2    1
32.4    1
35.1    1
Name: count, Length: 643, dtype: int64
feelslikemin
64.9    36
56.3    27
66.8    24
38.2    23
30.1    22
    ..
57.9    1
73.1    1
74.2    1
14.4    1
-16.6    1
Name: count, Length: 619, dtype: int64
feelslike
70.6    24
64.6    24
71.2    23
73.6    22
60.2    21
    ..
32.8    1
7.9     1
47.5    1
6.4     1
0.6     1
Name: count, Length: 657, dtype: int64
dew
37.1    21

```

```

27.3    20
22.1    20
55.1    19
59.8    18
..
10.0     1
31.7     1
55.6     1
52.4     1
32.2     1
Name: count, Length: 590, dtype: int64
humidity
71.4     31
77.6     28
73.2     27
71.1     26
65.4     24
..
35.4     1
39.7     1
54.2     1
47.7     1
99.2     1
Name: count, Length: 538, dtype: int64
precip
0.000    1640
0.001     86
0.008     75
0.002     49
0.010     39
...
0.763     1
0.436     1
1.267     1
1.447     1
1.526     1
Name: count, Length: 423, dtype: int64
precipcover
0.00    1640
4.17    300
8.33    235
20.83   184
12.50   179
16.67   178
29.17   129
25.00   123
37.50   102
41.67    90

```

33.33	82
45.83	76
50.00	64
54.17	61
66.67	37
58.33	36
62.50	33
75.00	28
70.83	21
100.00	20
91.67	16
87.50	16
83.33	13
79.17	11
95.83	10

Name: count, dtype: int64

snow

0.0	3396
0.2	72
0.1	63
0.3	39
0.4	19
0.5	16
0.6	13
0.7	10
1.1	8
0.9	8
1.2	6
1.0	6
1.6	5
0.8	5
1.4	4
1.5	4
2.7	3
1.3	2
2.9	2
2.1	2
1.9	1

Name: count, dtype: int64

snowdepth

0.0	3186
0.1	169
0.2	99
0.4	44
0.3	41
0.5	31
0.6	28
0.7	20

```

0.8      17
0.9      15
1.1       6
1.4       6
1.3       6
1.0       4
1.5       4
1.6       4
2.4       3
1.9       1
Name: count, dtype: int64
cloudcover
100.0     137
99.1      24
34.7      19
60.2      18
99.7      16
...
71.4       1
79.3       1
98.9       1
64.5       1
22.7       1
Name: count, Length: 803, dtype: int64
uvindex
4      540
5      498
8      450
6      415
2      407
1      401
7      378
3      375
9      182
0       20
10      18
Name: count, dtype: int64
severerisk
0.0      2428
10.0     1072
30.0      123
60.0       36
75.0       18
100.0        7
Name: count, dtype: int64
num_date
110      23
98       22

```

```

105    22
216    21
169    20
    ..
231     3
362     3
233     2
366     2
226     1
Name: count, Length: 366, dtype: int64
preciptype_freezingrain
0    3675
1         9
Name: count, dtype: int64
preciptype_freezingrain,snow
0    3674
1         10
Name: count, dtype: int64
preciptype_freezingrain,snow,ice
0    3680
1         4
Name: count, dtype: int64
preciptype_none
0    2311
1    1373
Name: count, dtype: int64
preciptype_rain
0    1923
1    1761
Name: count, dtype: int64
preciptype_rain,freezingrain
0    3672
1         12
Name: count, dtype: int64
preciptype_rain,freezingrain,snow
0    3680
1         4
Name: count, dtype: int64
preciptype_rain,snow
0    3305
1     379
Name: count, dtype: int64
preciptype_rain,snow,ice
0    3678
1         6
Name: count, dtype: int64
preciptype_snow
0    3558

```

```
1      126
Name: count, dtype: int64
```

6 Pair plot

Has both histogram distribution as well as general correlation information via shape for all features vs features.

```
[ ]: # sns.set_theme(style='whitegrid', palette='muted')
      # pp = sns.pairplot(data, diag_kind='kde')
      # pp.map_lower(sns.kdeplot, levels=4, color='.2')
```

```
[ ]: data.iloc[:, :16]
```

```
[ ]:      tempmax  tempmin  temp  feelslikemax  feelslikemin  feelslike  dew  \
0         26.2    -4.2  10.1         18.1         -16.6         0.6  -3.8
1         35.8    28.0  32.5         34.7          24.1        26.8  26.6
2         15.3     2.5   9.2         14.6          -6.0         3.5   0.5
3         15.3     2.5   9.2         14.6          -6.0         3.5   0.5
4         15.3     2.5   9.2         14.6          -6.0         3.5   0.5
...      ...      ...  ...      ...      ...      ...      ...
3679      68.8     58.9  61.6         68.8         58.9        61.6  56.7
3680      59.4     46.8  50.8         59.4         39.8        46.5  46.3
3681      59.2     44.5  51.1         59.2         37.4        47.9  33.0
3682      81.1     45.0  64.1         79.9         39.9        62.9  39.5
3683      81.1     45.0  64.1         79.9         39.9        62.9  39.5
```

```
      humidity  precip  precipcover  snow  snowdepth  cloudcover  uvindex  \
0         57.2    0.000         0.00   0.0         0.1         43.9         2
1         78.9    0.000         0.00   0.0         0.1         66.3         2
2         68.0    0.000         0.00   0.0         2.4         20.4         2
3         68.0    0.000         0.00   0.0         2.4         20.4         2
4         68.0    0.000         0.00   0.0         2.4         20.4         2
...      ...      ...      ...      ...      ...      ...      ...
3679      84.8    1.526        83.33   0.0         0.0         95.1         3
3680      84.5    0.266        83.33   0.0         0.0        100.0         2
3681      51.5    0.019        12.50   0.0         0.0         54.5         9
3682      42.6    0.001         4.17   0.0         0.0         25.2         8
3683      42.6    0.001         4.17   0.0         0.0         25.2         8
```

```
      severerisk  num_date
0             0.0         7
1             0.0         9
2             0.0        14
3             0.0        14
4             0.0        14
...      ...      ...
```

3679	10.0	102
3680	10.0	103
3681	10.0	104
3682	30.0	105
3683	30.0	105

[3684 rows x 16 columns]

```
[ ]: sns.set_theme(style='whitegrid', palette='muted')
pp = sns.pairplot(data.iloc[:, :16], diag_kind='kde')
pp.map_lower(sns.kdeplot, levels=4, color='.2')
```

```
c:\Users\btb51\anaconda3\envs\opioid_env\Lib\site-
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```

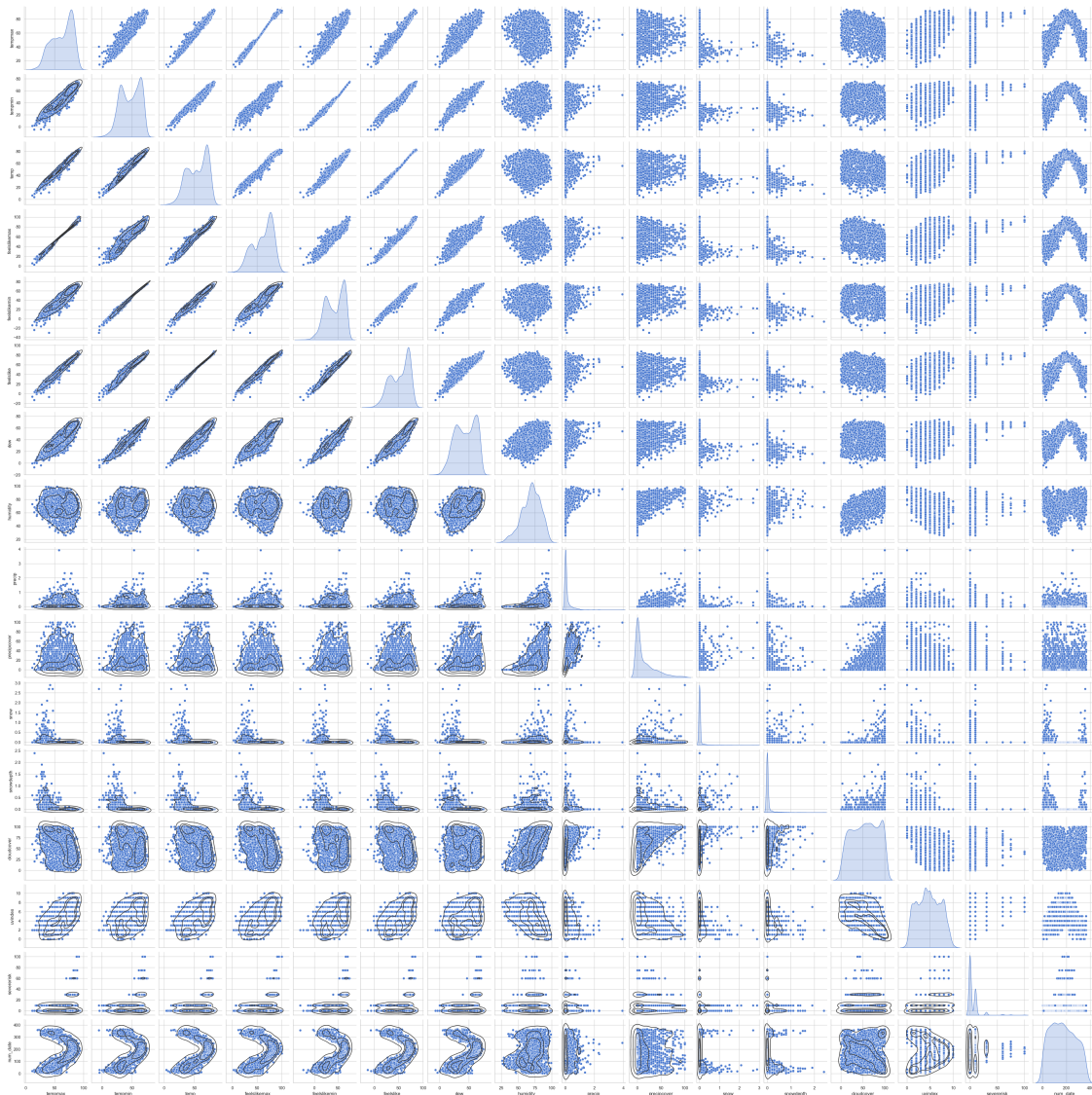
```
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```

```
with pd.option_context('mode.use_inf_as_na', True):
```

```
[ ]: <seaborn.axisgrid.PairGrid at 0x1fc9d9720c0>
```



7 Looking for any outliers in the dataset

Uses z-score of 3 to detect outliers

```
[ ]: cols
```

```
[ ]: Index(['tempmax', 'tempmin', 'temp', 'feelslikemax', 'feelslikemin',
            'feelslike', 'dew', 'humidity', 'precip', 'precipcover', 'snow',
            'snowdepth', 'cloudcover', 'uvindex', 'severerisk', 'num_date',
            'preciptype_freezingrain', 'preciptype_freezingrain,snow',
            'preciptype_freezingrain,snow,ice', 'preciptype_none',
            'preciptype_rain', 'preciptype_rain,freezingrain',
```

```

        'preciptype_rain,freezingrain,snow', 'preciptype_rain,snow',
        'preciptype_rain,snow,ice', 'preciptype_snow'],
        dtype='object')

```

```

[ ]: cols = data.columns
     num_cols = cols[:16]
     num_cols

```

```

[ ]: Index(['tempmax', 'tempmin', 'temp', 'feelslikemax', 'feelslikemin',
           'feelslike', 'dew', 'humidity', 'precip', 'precipcover', 'snow',
           'snowdepth', 'cloudcover', 'uvindex', 'severerisk', 'num_date'],
          dtype='object')

```

```

[ ]: # Old interim data:
     # num_cols = ['tempmax', 'tempmin', 'temp', 'feelslikemax',
     #             'feelslikemin', 'feelslike', 'dew', 'humidity', 'precip',
     #             ↪ 'precipcover',
     #             'snowdepth', 'cloudcover', 'uvindex', 'severerisk', 'moonphase']

     # New processed data:
     # num_cols = []

     # Z-scores
     for col in num_cols:
         data[col+'_z_score'] = (data[col] - data[col].mean()) / data[col].std()

     # thresh = 3 # Three std away for an outlier
     # outliers = data[np.abs(data['z_score']) > thresh]
     # outliers
     # There are no outliers

```

```

[ ]: data.iloc[:, 27:]

```

```

[ ]:
     tempmaxz_score  tempminz_score  tempz_score  feelslikemaxz_score \
0          -2.071784        -3.155281        -2.680809          -2.199752
1          -1.533162        -1.159831        -1.333731          -1.372516
2          -2.683345        -2.740079        -2.734932          -2.374170
3          -2.683345        -2.740079        -2.734932          -2.374170
4          -2.683345        -2.740079        -2.734932          -2.374170
...          ...          ...          ...          ...
3679          0.318353          0.755058          0.416267          0.326806
3680         -0.209048          0.005215         -0.233217         -0.141629
3681         -0.220270         -0.137317         -0.215176         -0.151596
3682          1.008462         -0.106332          0.566610          0.879957
3683          1.008462         -0.106332          0.566610          0.879957

     feelslikeminz_score  feelslikez_score  dewz_score  humidityz_score \

```

0	-3.053074	-2.664301	-2.759812	-0.871811
1	-0.980994	-1.320524	-0.991232	0.695221
2	-2.513417	-2.515563	-2.509651	-0.091906
3	-2.513417	-2.515563	-2.509651	-0.091906
4	-2.513417	-2.515563	-2.509651	-0.091906
...	...	...	...	...
3679	0.790711	0.464340	0.759895	1.121280
3680	-0.181691	-0.310127	0.154854	1.099616
3681	-0.303877	-0.238322	-0.618899	-1.283427
3682	-0.176599	0.531016	-0.240749	-1.926127
3683	-0.176599	0.531016	-0.240749	-1.926127

	precipz_score	precipcoverz_score	snowz_score	snowdepthz_score \
0	-0.431176	-0.714888	-0.194792	0.259506
1	-0.431176	-0.714888	-0.194792	0.259506
2	-0.431176	-0.714888	-0.194792	12.468556
3	-0.431176	-0.714888	-0.194792	12.468556
4	-0.431176	-0.714888	-0.194792	12.468556
...	...	...	...	...
3679	4.714085	3.110118	-0.194792	-0.271322
3680	0.465704	3.110118	-0.194792	-0.271322
3681	-0.367113	-0.141114	-0.194792	-0.271322
3682	-0.427804	-0.523477	-0.194792	-0.271322
3683	-0.427804	-0.523477	-0.194792	-0.271322

	cloudcoverz_score	uvindexz_score	severeriskz_score	num_datez_score
0	-0.386696	-1.136663	-0.470707	-1.626974
1	0.382002	-1.136663	-0.470707	-1.607234
2	-1.193143	-1.136663	-0.470707	-1.557886
3	-1.193143	-1.136663	-0.470707	-1.557886
4	-1.193143	-1.136663	-0.470707	-1.557886
...	...	...	...	...
3679	1.370327	-0.725159	0.460595	-0.689355
3680	1.538480	-1.136663	0.460595	-0.679485
3681	-0.022938	1.743866	0.460595	-0.669616
3682	-1.028422	1.332362	2.323201	-0.659746
3683	-1.028422	1.332362	2.323201	-0.659746

[3684 rows x 16 columns]

```
[ ]: z_values = data.iloc[:, 27:]
```

```
[ ]: z_values.max()
```

```
[ ]: tempmin_z_score      1.783768
      temp_z_score        1.721248
      feelslikemax_z_score 1.911511
```

```

feelslikemin_z_score    1.635834
feelslike_z_score       1.777344
dew_z_score             1.795445
humidity_z_score        2.161154
precip_z_score          12.792752
precipcover_z_score     3.875302
snow_z_score            14.863665
snowdepth_z_score       12.468556
cloudcover_z_score      1.538480
uvindex_z_score         2.155370
severerisk_z_score      8.842319
num_date_z_score        1.916237
dtype: float64

```

```
[ ]: z_values.min()
```

```

[ ]: tempmin_z_score    -3.204857
temp_z_score           -3.059674
feelslikemax_z_score   -2.967188
feelslikemin_z_score   -3.755646
feelslike_z_score      -3.372092
dew_z_score            -2.951796
humidity_z_score       -3.124870
precip_z_score         -0.431176
precipcover_z_score    -0.714888
snow_z_score           -0.194792
snowdepth_z_score      -0.271322
cloudcover_z_score     -1.893207
uvindex_z_score        -1.959671
severerisk_z_score     -0.470707
num_date_z_score       -1.686192
dtype: float64

```

```

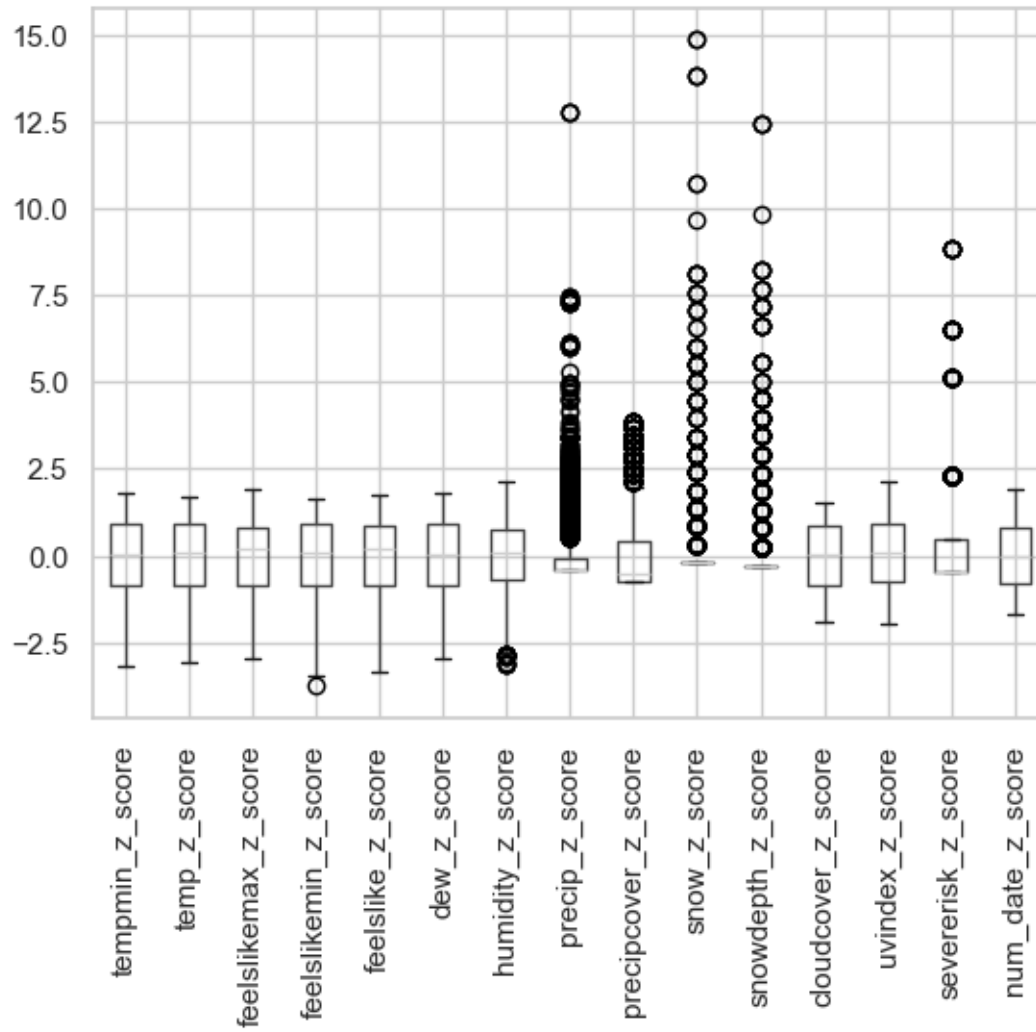
[ ]: tt = z_values.describe()
tt.to_csv('../data/processed/z_values_fivenumbersummary.csv')

```

```

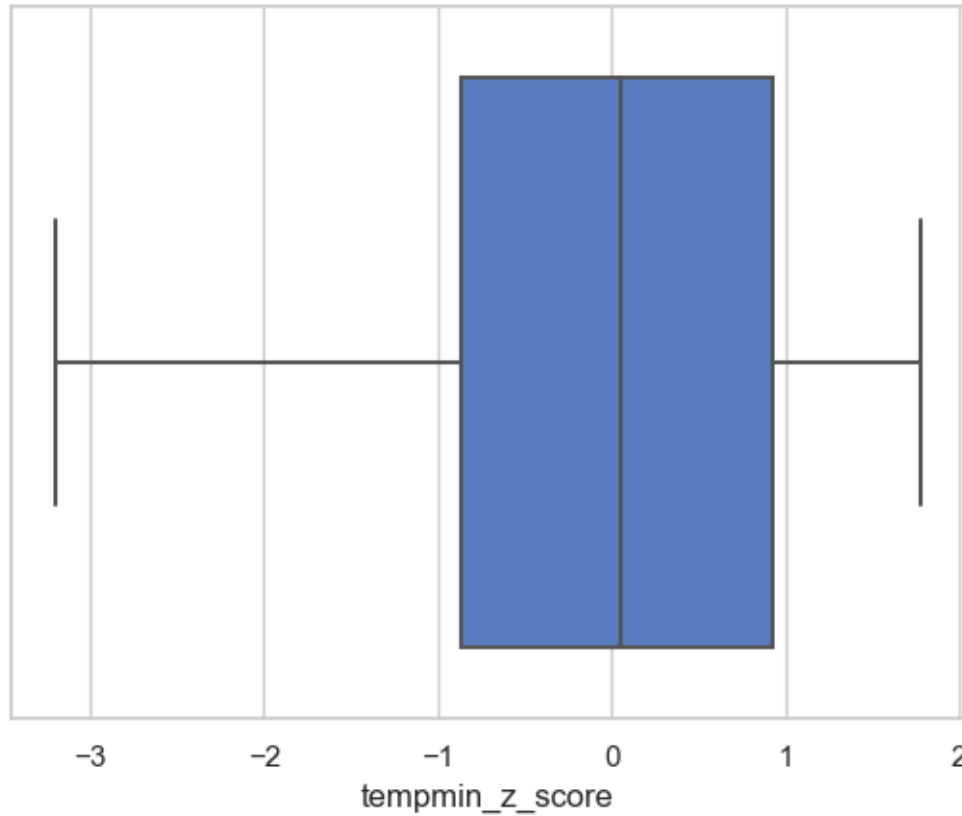
[ ]: z_values.boxplot()
plt.xticks(rotation=90)
plt.show()

```



```
[ ]: sns.boxplot(x='tempmin_z_score', data=z_values)
```

```
[ ]: <Axes: xlabel='tempmin_z_score'>
```



```
[ ]: z_values.describe()
```

```
[ ]:      tempmin_z_score  temp_z_score  feelslikemax_z_score  \
count      3.684000e+03  3.684000e+03      3.684000e+03
mean       3.548856e-16  9.257886e-17     -1.157236e-16
std        1.000000e+00  1.000000e+00      1.000000e+00
min        -3.204857e+00 -3.059674e+00     -2.967188e+00
25%        -8.623724e-01 -8.526319e-01     -8.492647e-01
50%         4.859402e-02  1.035525e-01      2.271387e-01
75%         9.239274e-01  9.033798e-01      8.151739e-01
max         1.783768e+00  1.721248e+00      1.911511e+00

      feelslikemin_z_score  feelslike_z_score  dew_z_score  \
count      3.684000e+03      3.684000e+03  3.684000e+03
mean       -1.697279e-16      2.314471e-16 -1.542981e-16
std        1.000000e+00      1.000000e+00  1.000000e+00
min        -3.755646e+00     -3.372092e+00 -2.951796e+00
25%        -8.740808e-01     -8.640504e-01 -8.457896e-01
50%         1.085025e-01      1.771204e-01  3.268270e-02
75%         9.294432e-01      8.797825e-01  9.111550e-01
max         1.635834e+00      1.777344e+00  1.795445e+00
```

	humidity_z_score	precip_z_score	precipcover_z_score	snow_z_score \
count	3.684000e+03	3.684000e+03	3.684000e+03	3.684000e+03
mean	3.703154e-16	-6.943414e-17	-7.136287e-17	-7.714905e-18
std	1.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00
min	-3.124870e+00	-4.311761e-01	-7.148878e-01	-1.947919e-01
25%	-6.641971e-01	-4.311761e-01	-7.148878e-01	-1.947919e-01
50%	8.501742e-02	-4.143174e-01	-5.234769e-01	-1.947919e-01
75%	7.529918e-01	-5.859984e-02	4.326597e-01	-1.947919e-01
max	2.161154e+00	1.279275e+01	3.875302e+00	1.486366e+01

	snowdepth_z_score	cloudcover_z_score	uvindex_z_score \
count	3.684000e+03	3.684000e+03	3.684000e+03
mean	1.542981e-17	-2.005875e-16	-1.465832e-16
std	1.000000e+00	1.000000e+00	1.000000e+00
min	-2.713218e-01	-1.893207e+00	-1.959671e+00
25%	-2.713218e-01	-8.576954e-01	-7.251588e-01
50%	-2.713218e-01	2.510609e-02	9.784952e-02
75%	-2.713218e-01	8.761645e-01	9.208578e-01
max	1.246856e+01	1.538480e+00	2.155370e+00

	severerisk_z_score	num_date_z_score
count	3.684000e+03	3.684000e+03
mean	-6.171924e-17	-3.085962e-17
std	1.000000e+00	1.000000e+00
min	-4.707073e-01	-1.686192e+00
25%	-4.707073e-01	-8.201282e-01
50%	-4.707073e-01	-4.782664e-02
75%	4.605954e-01	8.108346e-01
max	8.842319e+00	1.916237e+00

```
[ ]: condition = (z_values > 3.0) | (z_values < -3.0)
      condition
```

```
[ ]: tempmin_z_score temp_z_score feelslikemax_z_score \
0      True      False      False
1      False     False     False
2      False     False     False
3      False     False     False
4      False     False     False
...
3679   False     False     False
3680   False     False     False
3681   False     False     False
3682   False     False     False
3683   False     False     False
```

	feelslikemin_z_score	feelslike_z_score	dew_z_score	humidity_z_score	\
0	True	False	False	False	
1	False	False	False	False	
2	False	False	False	False	
3	False	False	False	False	
4	False	False	False	False	
...	...	...	...	...	
3679	False	False	False	False	
3680	False	False	False	False	
3681	False	False	False	False	
3682	False	False	False	False	
3683	False	False	False	False	

	precip_z_score	precipcover_z_score	snow_z_score	snowdepth_z_score	\
0	False	False	False	False	
1	False	False	False	False	
2	False	False	False	True	
3	False	False	False	True	
4	False	False	False	True	
...	...	...	...	...	
3679	True	True	False	False	
3680	False	True	False	False	
3681	False	False	False	False	
3682	False	False	False	False	
3683	False	False	False	False	

	cloudcover_z_score	uvindex_z_score	severerisk_z_score	\
0	False	False	False	
1	False	False	False	
2	False	False	False	
3	False	False	False	
4	False	False	False	
...	...	...	...	
3679	False	False	False	
3680	False	False	False	
3681	False	False	False	
3682	False	False	False	
3683	False	False	False	

	num_date_z_score
0	False
1	False
2	False
3	False
4	False
...	...
3679	False

```

3680          False
3681          False
3682          False
3683          False

```

```
[3684 rows x 15 columns]
```

```
[ ]: rows_with_condition = condition.any(axis=1)
rows_with_condition
```

```

[ ]: 0          True
      1          False
      2          True
      3          True
      4          True
      ...
3679    True
3680    True
3681   False
3682   False
3683   False
Length: 3684, dtype: bool

```

```
[ ]: sum(rows_with_condition)
```

```
[ ]: 318
```

```
[ ]: bad_index = z_values[rows_with_condition].index
bad_index
```

```

[ ]: Index([ 0,  2,  3,  4, 11, 13, 40, 41, 42, 63,
            ...
            3627, 3645, 3651, 3652, 3667, 3668, 3669, 3670, 3679, 3680],
           dtype='int64', length=318)

```

```
[ ]: data2 = data.drop(index=bad_index)
data2 = data2.iloc[:, 0:26]
data2
```

```

[ ]:   tempmax  tempmin  temp  feelslikemax  feelslikemin  feelslike  dew  \
1      35.8    28.0  32.5      34.7      24.1    26.8  26.6
5      55.1    35.8  43.6      55.1      25.6    37.1  34.5
6      47.2    35.0  41.2      44.7      31.4    38.9  30.6
7      42.9    28.1  34.7      41.1      18.5    31.9  25.3
8      44.2    27.1  40.2      37.6      16.0    33.2  27.5
...      ...      ...      ...      ...      ...      ...
3677    73.0    49.3  60.1      73.0      47.3    60.0  37.1
3678    65.1    57.1  60.6      65.1      57.1    60.6  54.5

```

3681	59.2	44.5	51.1	59.2	37.4	47.9	33.0
3682	81.1	45.0	64.1	79.9	39.9	62.9	39.5
3683	81.1	45.0	64.1	79.9	39.9	62.9	39.5

	humidity	precip	precipcover	...	preciptype_freezingrain	\
1	78.9	0.000	0.00	...	0	
5	70.8	0.154	29.17	...	0	
6	69.3	0.052	8.33	...	0	
7	69.6	0.035	16.67	...	0	
8	62.9	0.090	25.00	...	0	
...	...	...	...	...	...	
3677	42.8	0.006	8.33	...	0	
3678	80.9	0.179	62.50	...	0	
3681	51.5	0.019	12.50	...	0	
3682	42.6	0.001	4.17	...	0	
3683	42.6	0.001	4.17	...	0	

	preciptype_freezingrain,snow	preciptype_freezingrain,snow,ice	\
1	0	0	
5	0	0	
6	0	0	
7	0	0	
8	0	0	
...	...	...	
3677	0	0	
3678	0	0	
3681	0	0	
3682	0	0	
3683	0	0	

	preciptype_none	preciptype_rain	preciptype_rain,freezingrain	\
1	1	0	0	
5	0	1	0	
6	0	1	0	
7	0	0	0	
8	0	0	0	
...	...	...	...	
3677	0	1	0	
3678	0	1	0	
3681	0	1	0	
3682	0	1	0	
3683	0	1	0	

	preciptype_rain,freezingrain,snow	preciptype_rain,snow	\
1	0	0	
5	0	0	
6	0	0	

7	0	1
8	0	1
...	...	...
3677	0	0
3678	0	0
3681	0	0
3682	0	0
3683	0	0

	preciptype_rain,snow,ice	preciptype_snow
1	0	0
5	0	0
6	0	0
7	0	0
8	0	0
...	...	...
3677	0	0
3678	0	0
3681	0	0
3682	0	0
3683	0	0

[3366 rows x 26 columns]

```
[ ]: data2.to_csv('../data/processed/cleaned_data_no_outliers.csv')
```

```
[ ]: z_values.to_csv('../data/processed/z_valuesforoutliers.csv')
```